Fresh Shuffles, Reused Secrets: Breaking KV-Cloak with Repeated-Token Prompts

***Abstract*—We show how input symmetry can neutralize fresh shuffling in KV-Cloak, an NDSS 2026 defense that combines secret linear transformations, additive markers, and independently shuffled KV-cache blocks. A repeated-token text prompt makes the first-layer Value rows identical. Shuffling then changes only the marker's position: protected blocks occupy an orbit of at most b states, despite $b!$ possible permutations. Rank-one differences between these states reveal an equivalent row demixer and the implementation's structured column transform. The recovered parameters transfer to other requests protected by the same secrets, exposing token multisets before optional language-prior ordering. We characterize this orbit, establish identification under explicit nondegeneracy conditions, and derive a query bound. Against the official implementation, the attack recovers 263,029 of 263,904 head–token evaluation instances across ten models and twelve configurations. All fifteen additional model–key epochs calibrate successfully, with model-level mean recovery between 99.34% and 100%. A synthetic ordering follow-up logs 22.3% mean position accuracy over 600 records versus 7.4% randomized ordering; repeated texts and nominal-length overshoots limit interpretation. A limited study of 30 synthetic closed-set values under one key yields 77.36% expected top-1 identification under uniform tie-breaking; this is not a population-level privacy estimate. Request-specific refresh of the row transform and marker blocks the tested cross-request transfer, without establishing a complete secure repair. These results show why fresh permutations can leave reused secrets exposed when input symmetry reduces their observable effect to a small marker orbit.**

1. Introduction

The KV cache stores the history of a language-model interaction as reusable inference state. A party that observes this state may recover information about the prompt even without access to the client connection. Cache offloading and reuse therefore raise a confidentiality question alongside their performance benefits. Block-oriented management, exemplified by PagedAttention [1], makes small cache blocks a natural unit for both protection and analysis.

KV-Cloak [2] addresses this question with reversible obfuscation. Its implementation combines a secret column transform, a sparse additive marker, a fresh row permutation for each block, and a secret row transform. The column transform can be folded into attention weights; the other operations protect stored blocks. The design's security argument assigns an important role to the $b!$ possible row permutations. In the published paper, fresh shuffling is the step intended to defeat recovery of a fixed linear transform through chosen plaintexts. This suggests a focused question: does the number of permutations describe the uncertainty that the attacker actually observes?

Our answer is negative for the tested construction. A permutation changes the ordering of a block, but cannot change a block whose rows are all identical. Such blocks are reachable without selecting arbitrary hidden vectors. In the first Value layer, each row is a deterministic function of its input token under the model architectures we evaluate. Repeated characters can tokenize to a long run of the same token. An attacker submits the text through the normal tokenizer and observes its protected cache. Mandatory prefixes are retained; the attacker selects complete blocks inside the repeat run after observing the cache.

The resulting protected blocks vary only with the shuffled position of the additive marker. For a single-row marker, there are b ideal states, rather than $b!$. Their pairwise differences have rank one. Those differences expose the marker direction, and the geometry of the states exposes the columns of the reused row transform. A known probe vector then identifies the implementation's independent two-dimensional column rotations, subject to explicit nondegeneracy conditions. The attacker can apply the recovered parameters to unrelated blocks from the same secret epoch and match the resulting vectors against a dictionary computed from the public, original model.

This is a confidentiality attack under a cache-observation and chosen-plaintext threat model. It is not a method for obtaining cache access from an ordinary text API. The algebraic output is a token multiset for each block. We separately evaluate public-model ordering of these bags: 600 synthetic records yield 22.3% mean position accuracy and 18 exact evaluated spans, with repeated inputs and length-accounting limitations. Content recovery, language-prior sequence estimates, and faithful factual reconstruction therefore remain distinct claims.

The evidence spans several complementary settings. Our main suite uses ten models and twelve configurations, including tokenizer prefixes, different head dimensions, and block sizes 8, 16, and 32. It records raw correct/total counts rather than reconstructing them from rounded rates. A separate test attacks cache files produced by the official command-line entry point for 16 LMSYS conversations. An isolated fused-mode experiment constructs the attack dictionary from the original public model, not the defender's

fused weights. A further three-model study varies complete secret configurations over five epochs per model. These experiments support the mechanism while leaving deployment prevalence and unseen-conversation generalization unmeasured.

We also examine what the recovered contents mean for privacy and mitigation. A closed-set synthetic study ranks 100 same-type value candidates in each of 30 cases. The true candidate always attains the maximum score, but ties remain: only 21 cases have a unique maximum. This is a bounded observation, with a fixed candidate-selection protocol and incomplete retained traces, not a general identification-rate claim. Finally, a request-level refresh of the row transform and marker prevents the existing recovered demixer from transferring in one measured setting. This simple intervention also narrows the research conclusion: the attack concerns reused structural secrets, and a new obfuscation system is not needed to state that finding.

As a bounded follow-up, VeilKV-Plus combines authenticated byte protection with an explicit public-prefix/private-tail policy. Its measured advantages concern cache-protocol integrity, stored-prefix reuse, and operation cost relative to the tested KV-Cloak baseline under a simulated TEE boundary. A separate arithmetic-profile study finds closed-set spectral identification above the candidate baseline, so that profile is excluded from confidentiality claims. Section 6.4 states the baseline comparisons and their distinct assumptions.

Our contributions are:

- We characterize the observable orbit induced by chosen-input symmetry in shuffled cache obfuscation, deriving state counts, identification conditions, and a collection bound. The analysis distinguishes the single-marker result from more general marker patterns.
- We construct a repeated-token attack on the official KV-Cloak implementation that recovers an equivalent first-layer Value demixer and transfers it across requests without accessing the defender's secret parameters or fused dictionary.
- We evaluate recovery, calibration cost, secret-epoch stability, closed-set value identification, and a simple refresh intervention, distinguishing raw-count experiments, legacy summaries, and unresolved properties.

2. System and Threat Model

2.1. Cache representation and the tested transform

A decoder stores Key and Value vectors for previously processed tokens. For the first Value layer of the evaluated models, write

$$v(t) = W_V\, N(E[t]) + \beta_V, \tag{1}$$

where E is the public embedding table, N is the first layer's input normalization, and W_V, β_V are the relevant head's projection and bias. The implementation detects the appropriate normalization/projection path for each model. Equation (1) does not assert that deeper-layer states depend only on the token. It also does not apply to every possible architecture with absolute position-dependent inputs.

Let $V \in \mathbb{R}^{b\times d}$ contain one head's b Value rows. The official default obfuscation path can be written as

$$C = SP(VM + A), \qquad A = e_r a^{\mathsf{T}}. \tag{2}$$

Here $S \in O(b)$ is a secret orthogonal row transform, $M \in O(d)$ is a structured column transform, P is a block-specific permutation, and A adds a secret vector a to one row. The marker vector is positive in the evaluated configuration. For a fixed head and branch, S, M, and a are reused by a cloak instance. The marker's effective location changes under P. The implementation also exposes optional positive scaling factors; our exact derivation first addresses the default orthogonal configuration, and we evaluate one scaled configuration separately.

Equation (2) follows the implementation's operation order: A is added after the column transform and is expressed in transformed-Value coordinates. The implementation pairs coordinate i with $i + d/2$ in M. Each pair is rotated independently. This structure is consequential: recovering an arbitrary d-dimensional orthogonal matrix from one vector pair would be underdetermined. We do not make that claim. Rotary position embeddings motivate structured transforms in the original design [2], [3]; the implementation uses the paired structure on the Value branch as well.

2.2. Attacker capabilities

The adversary knows the original public model and tokenizer. It can submit its own text requests and read the corresponding protected cache blocks. It can also observe a target request's protected cache under the same relevant secret configuration. This observation may model an untrusted cache reader or a cache disclosure, but obtaining that access is outside our attack. A successful experiment does not show that an arbitrary remote API client can read another user's cache.

The adversary does not receive S, M, a, victim plaintext, or the defender's fused model weights. A public local model supplies the dictionary and the plaintext vector for the attacker's own probe. Defender secrets may be used by evaluation-only diagnostics, but are not inputs to the recovery routine. The isolated fused experiment reloads the original model for the attacker and computes its transformed dictionary using recovered parameters.

We assume public block size and head layout. The adversary may calibrate and attack only within a secret epoch shared by probe and target. Instance routing, per-tenant key scopes, or request-specific transformations can prevent that sharing. We do not establish how often real deployments satisfy this condition. The measured lifecycle experiment is a controlled simulation of the shared-secret setting.

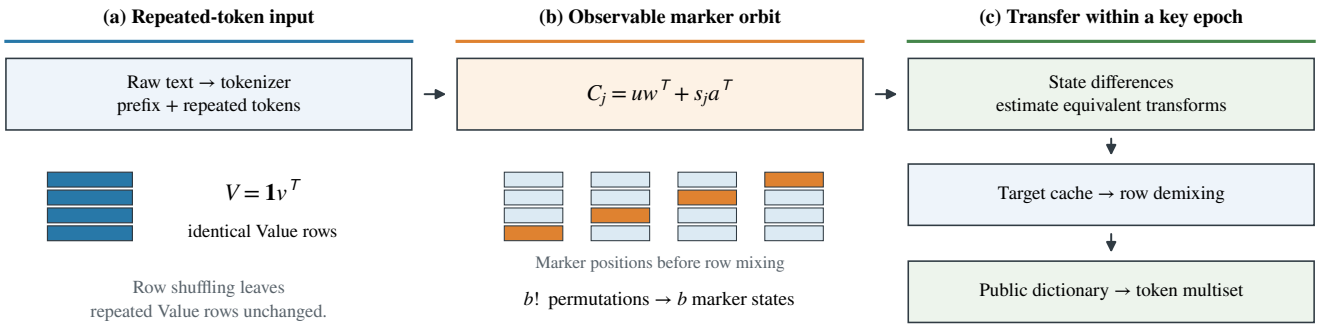


Figure 1. Attack mechanism. A legal repeat run makes first-layer Value rows identical. Fresh permutations move only the marker, producing a small orbit. The attacker recovers equivalent transforms from its own protected blocks, then matches demixed victim rows against a public-model dictionary. The diagram is schematic; no victim plaintext or server secret is an attack input.

The relevant secret scope is an implementation instance, rather than the mere fact that two requests use the same model. Two copies of a public checkpoint can have different cloak configurations. Conversely, repeated requests processed by one configured instance can share the row and column transforms even though their block permutations differ. Our target condition is the latter: the adversary's probe and the observed target cache use the same relevant secrets. This condition belongs to the threat model and is not inferred from the model identifier.

It is useful to distinguish two boundaries in assessing that condition. The first is access to cache bytes, which may be governed by a storage or execution boundary. The second is the confidentiality supplied by the cloak after those bytes become visible. Our analysis addresses the second boundary. A deployment that prevents cross-request cache observation removes a required capability; a deployment that permits observation must evaluate what its transform conceals. The controlled experiments instantiate this observation capability explicitly and therefore do not measure how frequently either deployment condition holds.

2.3. Goals and exclusions

The primary goal is recovery of the token multiset in each complete target block. A secondary goal is identifying a synthetic value from an explicitly supplied same-type candidate set. The second goal is conditional on that candidate knowledge; it is not unconstrained recovery of an arbitrary credential.

The algebraic output does not assign within-block positions; the language-prior follow-up in Section 5.3 estimates sequences separately. We do not establish general exact-prompt reconstruction, deeper-layer recovery, or inference on an untrusted accelerator with unrestricted access to all intermediate computation. We also distinguish floating-point precision: model forward passes are FP32 in the main suite, while cloak matrix operations use bf16. This is not an end-to-end bf16 serving benchmark.

TABLE I. INFORMATION BOUNDARY USED BY THE ATTACK.

Available to attacker	**Evaluation only / unavailable**
Original model and tokenizer	Secret configuration
Own text and protected probe	Victim plaintext and token labels
Protected target blocks	Defender's fused dictionary
Public block/head dimensions	True per-block permutation
Supplied candidates (value task)	True candidate index for scoring

3. From Permutation Count to Observable Orbit

The analysis separates three questions: how many states can a chosen input produce; whether those states identify reusable parameters; and how many blocks are required to observe them. A large permutation group alone answers none of these questions.

3.1. Input symmetry collapses the orbit

Suppose a legal input produces $V = \mathbf{1}v^{\mathsf{T}}$. Every row permutation then satisfies $PV = V$. Define $w = M^{\mathsf{T}}v$ and $u = S\mathbf{1}$. Equation (2) becomes

$$C(P) = uw^{\mathsf{T}} + SPA. \tag{3}$$

Only the marker moves. Identical rows, rather than rank one alone, are essential: a general rank-one matrix xv^{T} need not be invariant when the entries of x differ.

Proposition 1 (Marker orbit)**.** *For fixed $V = \mathbf{1}v^{\mathsf{T}}$, fixed A, M, and invertible S, the number of distinct exact-arithmetic ciphertexts under all row permutations is*

$$|\{C(P) : P \in \mathcal{S}_b\}| = \frac{b!}{|\mathrm{Stab}(A)|}, \tag{4}$$

where $\mathrm{Stab}(A) = \{P : PA = A\}$. For $A = e_r a^{\mathsf{T}}$ with $a \neq 0$, this number is b.

Proof. Adding the constant uw^{T} and multiplying by invertible S preserve equality of outputs. Thus two outputs agree exactly when their permuted markers agree. The orbit–stabilizer identity gives (4). A single nonzero marker row can occupy b distinct positions. □

For k identical nonzero marker rows and $b-k$ zero rows, the count is $\binom{b}{k}$. If all marker rows are identical, it is one even though the marker has full support. If the k nonzero rows are distinct and different from zero, the count is $b!/(b-k)!$. These examples prevent a misleading sparse-versus-dense dichotomy: marker support size alone does not determine security. Our implemented recovery routine uses the single-row case, not an attack on every orbit in (4).

In that case, let $s_j = Se_j$. The b ideal states are

$$C_j = uw^{\mathsf{T}} + s_j a^{\mathsf{T}}, \qquad j = 1, \ldots, b. \tag{5}$$

Fresh P remains present in the implementation. The attack neither predicts its random seed nor forces it to repeat. Instead, many permutations have the same effect on the chosen input.

3.2. Identification from the state geometry

The next result describes an equivalent demixer, not the original order of S's columns.

Theorem 2 (Default-configuration identification)**.** *Assume $b \geq 2$; all states in (5) are available; S is orthogonal; a is nonzero with known positive orientation; v is known; and w has a nonzero component orthogonal to a. Assume further that the known norm $\|w\| = \|v\|$ distinguishes the two signs of u. Then a, w, u and the columns of S are identifiable, with the columns unordered. If M consists of independent two-dimensional rotations and the restriction of v to every pair is nonzero, M is identifiable modulo full rotations.*

Proof. For $i \neq j$,

$$C_i - C_j = (s_i - s_j)a^{\mathsf{T}}. \tag{6}$$

This matrix has rank one. Its right singular direction identifies $a/\|a\|$ up to sign; positivity fixes the orientation. Since distinct orthonormal columns satisfy $\|s_i - s_j\| = \sqrt{2}$, its nonzero singular value is $\sqrt{2}\|a\|$, identifying the magnitude.

Let $Q_a = I - aa^{\mathsf{T}}/\|a\|^2$. Then $C_j Q_a = u(Q_a w)^{\mathsf{T}}$ is a nonzero rank-one matrix. Its left direction identifies u up to sign, and $\|u\| = \sqrt{b}$ fixes its magnitude. For either sign candidate, use $u^{\mathsf{T}} s_j = 1$ to compute

$$w^{\mathsf{T}} = \frac{u^{\mathsf{T}} C_j - a^{\mathsf{T}}}{b}. \tag{7}$$

The assumed norm separation selects the correct sign. Each column follows from

$$s_j = \frac{(C_j - uw^{\mathsf{T}})a}{\|a\|^2}. \tag{8}$$

The state labels do not identify the original column indices, leaving an arbitrary column permutation. Finally, within every nonzero coordinate pair, the angle from the known v pair to the recovered w pair identifies the corresponding rotation under the implementation's convention. □

The qualifications are substantive. Choosing $-u$ in (7) gives $w' = -w - 2a/b$, not merely a harmless sign flip. The norm test fails to distinguish signs precisely when $\langle w, a\rangle = -\|a\|^2/b$. Similarly, a zero pair in v cannot reveal that pair's rotation. A different legal probe may resolve these degeneracies, but our theorem does not promise that one arbitrary token always suffices. Nor does it extend one-pair identification to a general orthogonal M.

The identification result separates a loss of randomization from a recovery guarantee. A small orbit alone does not imply that the underlying transform is identifiable. Here, identification additionally uses orthogonal columns, the marker's known orientation, a public probe vector, and the restricted structure of M. Each assumption supplies a different missing piece: differences identify the marker, projection isolates the common component, and the public probe fixes the column rotation. The theorem does not infer these properties merely from observing b clusters.

Likewise, an equivalent demixer need not reveal the original secret representation. Relabeling the recovered columns changes their order but leaves the content-recovery objective unchanged. This distinction explains why the absence of original column labels is compatible with high multiset recovery, while it remains a material obstacle to claims about token positions. The downstream marker-selection heuristic and dictionary match still have to be evaluated after the algebraic identification step.

3.3. Collection cost and finite precision

If each block independently maps the marker uniformly to a row, collecting all b states is a coupon-collector process. The expected block count is bH_b, with $H_b = \sum_{i=1}^{b} 1/i$. For $b = 16$, this is about 54.1 blocks. A union bound gives

$$\Pr[\text{some state missing after } n] \leq b(1-1/b)^n \leq be^{-n/b}. \tag{9}$$

For H heads, $n \geq b\log(bH/\delta)$ suffices for the analogous union-bound failure probability δ. Heads need not be independent for this union bound; blockwise sampling within each head must satisfy the premise. Prefixes and padding reduce the number of usable blocks. A count of observed clusters is a calibration diagnostic, not a guarantee of success within a fixed budget or after the key has changed.

Floating-point implementations need not produce identical bit patterns for permutations in the same ideal state. Different accumulation orders can create nearby realizations. If every observed block is within Frobenius distance ϵ of its ideal state, within-state distances are at most 2ϵ. For distinct states, orthogonality gives

$$\|C_i - C_j\|_F = \sqrt{2}\|a\|. \tag{10}$$

Consequently a separating distance threshold exists whenever $4\epsilon < \sqrt{2}\|a\|$. This sufficient condition explains why rounding variants can remain clusterable; it is not an asserted

error bound for every hardware backend. The experiments measure recovery under the actual bf16 cloak operations instead of inferring it solely from ideal arithmetic.

For dictionary decoding, suppose the true normalized vector z has margin γ over every incorrect normalized dictionary vector in inner product, and the recovered normalized vector differs from z by at most η in Euclidean norm. Each score changes by at most η, so $2\eta < \gamma$ suffices to preserve its nearest neighbor. This conditional statement does not cover selecting the correct marker hypothesis, and it does not replace measurements of dictionary ambiguity. We make no uniform numerical-margin claim across the vocabulary.

4. Attack Construction

4.1. Reachable probes and state recovery

The attack searches a small character set for text whose normal tokenization contains a sufficiently long constant-token run. It submits the tokenizer's complete output, including any beginning-of-sequence or other prefix tokens. If the run begins at token offset p and the input contains L tokens, the usable block indices are

$$\lceil p/b \rceil, \ldots, \lfloor L/b \rfloor - 1. \tag{11}$$

This filtering happens on the attacker side. It does not remove the prefix before the model computes its cache. The main suite records the probe text character, prefix length, total token count, and whether this raw-entry path succeeded. All twelve included configurations use that path.

For each head, the attacker flattens protected probe blocks, groups nearby states, and computes a representative for each group. It estimates the rank-one right singular directions in (6), aligns their signs, and combines them to estimate a. A numerical approximation to the orthogonal projection then estimates u; the known public probe vector selects its sign. Equations (7) and (8) supply the remaining estimates. The implementation checks state coverage before attempting recovery. Its averaged marker direction is not renormalized before projection, and the left singular vectors are aggregated without explicit sign alignment. These are measured numerical heuristics, not a backend-independent implementation guarantee of Theorem 2. Ground-truth column similarities are evaluation diagnostics, not decisions supplied to the attack.

Let $\widetilde{S}$ contain the recovered columns in state order. In exact arithmetic, $\widetilde{S} = S\Pi$ for some unknown permutation Π. Then $\widetilde{S}^{\mathsf{T}} S = \Pi^{\mathsf{T}}$. This unordered recovery is sufficient for content extraction: it only changes the order in which recovered rows are returned.

4.2. Decoding an unseen block

For each protected target block C, the attacker enumerates the b possible marker columns. For hypothesis j, it forms

$$Y_j = \widetilde{S}^{\mathsf{T}}(C - \widetilde{s}_j a^{\mathsf{T}})\widehat{M}^{\mathsf{T}}. \tag{12}$$

Under the correct marker hypothesis and exact recovery, Y_j is a row-permuted copy of V. For each row, the attacker selects the nearest normalized vector from the dictionary in (1). It chooses the marker hypothesis with the largest sum of rowwise best-match cosine scores. This final selection is a measured heuristic; the identification theorem does not by itself guarantee its success for every vocabulary and block.

The output preserves multiplicity but not within-block order. We score it by multiset intersection:

$$\mathrm{hit}(\widehat{T}, T) = \sum_t \min\{n_{\widehat{T}}(t), n_T(t)\}. \tag{13}$$

The denominator is b per scored head and block. Repeated observations of the same token across heads remain separate evaluation opportunities; they are not independent textual samples.

The dictionary is a one-time public-model computation. Calibration uses state clustering and small-matrix factorizations. Target recovery tries b marker hypotheses and performs rowwise dictionary comparisons. Thus the attack does not enumerate $b!$ permutations. Its practical cost still depends on vocabulary size, dimension, and implementation of nearest-neighbor search. We do not present a deployment-throughput claim from the current CPU scripts.

The evaluation tracks calibration and content recovery separately. Observing too few states prevents the implemented calibrator from producing a head-specific demixer. Observing all states allows parameter estimation, but does not guarantee a correct marker hypothesis or a unique dictionary match on each target block. A successful calibration diagnostic therefore cannot stand in for the content metric. The main suite records both the state count per head and the target-text hit counts, allowing these two stages to be checked independently.

The public dictionary also has a precise scope. It represents the first Value projection of each vocabulary embedding with the model's input normalization and any projection bias. It is not a table of victim contexts, and its construction does not require victim labels. Different tokenizers, normalization paths, or projection layouts can change this dictionary; a model-family name alone is insufficient to select it. The implementation checks the dictionary against locally computed probe states. This check establishes consistency for those known inputs, rather than injectivity of the entire vocabulary representation.

4.3. Fusion, scaling, and secret scope

In fused mode, the server folds M into its projection weights, producing VM before the cache obfuscator. The attacker still uses its unfused public model. It recovers the corresponding transformed probe vector from protected states and composes the public dictionary with $\widehat{M}$. Our isolated experiment uses independent public and defender model objects; the earlier shortcut of constructing an attack dictionary from fused defender weights is excluded from its reported result.

The optional scaling configuration changes orthogonal matrices into structured rotation/scaling transforms. It requires modified recovery equations rather than an unqualified application of Theorem 2. We include its recorded behavior as a separate implementation variant. The exact theorem applies to its stated orthogonal setting only.

Parameters are recovered for a head and secret epoch. A new permutation does not invalidate them; a new secret transform may. This distinction is tested by applying recovered parameters to fresh requests under the same configuration, to a new configuration as a negative control, and to a refreshed configuration after recalibration. Those tests isolate parameter reuse from a claim that the attack survives arbitrary key rotation.

5. Evaluation

We ask four questions. **RQ1** tests whether legal repeated-token inputs produce the predicted small orbit and identify an equivalent demixer. **RQ2** measures token-multiset recovery across models and configurations, followed by language-prior ordering. **RQ3** tests the official entry point and the effect of secret lifetimes. **RQ4** asks whether the recovered content supports a concrete, bounded privacy task and whether shortening the secret lifetime interrupts transfer.

5.1. Experimental methodology

The main suite uses the official KV-Cloak implementation at commit `6b40f36`. We do not modify its cache transformation. Model forward passes use FP32 weights on CPU, and the cloak stores/transforms cache blocks in bf16, matching the implementation default. Experiments ran on an AMD EPYC 9554 host. Table II reports ten distinct public models from the Qwen, Llama, OLMo, Phi, and TinyLlama families [4], [5], [6], [7], [8] and twelve configurations; Qwen2.5-0.5B additionally covers block sizes 8 and 32. Checkpoint revisions are listed in Appendix A.

For every configuration, the attacker submits text through the unmodified tokenizer. Any beginning-of-sequence or other prefix token remains in the model input. Calibration uses only complete blocks inside the final constant-token run; content recovery is scored on complete blocks of the target texts. Calibration requires all b states to be observed for a head. Every head in the main suite calibrated, so Table II includes all heads.

Five target texts cover English prose, Python, Chinese, dialogue, and mathematical prose. Each is evaluated under two fresh block-permutation seeds. The primary metric is the multiset hit count in (13), summed over complete blocks and calibrated heads. Thus a "head–token opportunity" is one token position scored through one KV head. Opportunities within a document, across heads, and across seeds are correlated. We report their raw descriptive total and per-configuration rates; we do not use a token-level binomial interval as evidence of generalization to independent requests.

The main-suite fresh-secret control applies a demixer calibrated under one complete configuration to a cache produced under another, scoring head 0 on the English prose target. Its denominator therefore differs from the multi-head, five-text primary metric. Ground-truth secrets are used only for diagnostics such as parameter similarity. The attack routine receives the public model, its own protected calibration cache, and protected target blocks.

Configuration and input accounting. The main runner estimates the defender's magnitude thresholds from eight fixed natural-language sentences stored separately from the five target texts. It generates one secret configuration per model/block-size cell with seed 42, then evaluates two target-cache permutation seeds, 5001 and 5002. These two seeds change the block permutations; they do not constitute two independently generated secret configurations. The additional multi-epoch study below varies the complete configuration and answers a different question.

Probe lengths are recorded after tokenization. Character repetition can produce different token counts across tokenizers, and the search may overshoot its target length. The table therefore reports the actual submitted token count rather than a nominal character or token budget. Prefix lengths are one token for the recorded Llama and Phi configurations and two for TinyLlama; the remaining main-suite configurations have no prefix before the constant-token run. Prefix-containing blocks and incomplete final probe blocks are excluded from calibration. Target texts are independently truncated to their last complete block, so the reported recovery denominator excludes their trailing partial blocks.

The main suite counts head–token opportunities, the probe curve calibration trials, the epoch study session rates, and the privacy task candidate rankings. We keep these denominators separate.

5.2. RQ1: orbit and parameter recovery

Every evaluated head forms exactly b clusters after prefix blocks are removed: 8, 16, or 32 according to the configuration. The smallest saved between/within cluster-separation ratio is 500.2. Evaluation matches recovered columns of S to true columns by cosine similarity and checks that the matching is bijective. The minimum matched cosine similarity is 0.999996; the largest relative error recorded for w is 0.0354. These evaluation-only diagnostics support the predicted rank-one state geometry.

Figure 2 compares the observed two-head calibration rate on Qwen2.5-0.5B with the exact one-head occupancy probability for independent uniform states. Each empirical point contains ten trials under one fixed secret configuration. This diagnostic counts a trial as successful when both heads collect all states and the recorded marker and minimum matched-column cosine similarities each reach 0.999. With 80 blocks (1,280 repeated tokens), 9/10 trials meet this criterion; with 128 blocks (2,048 tokens), all ten do. The small, independently seeded trial sets can yield a non-monotone empirical curve. The comparison supports the

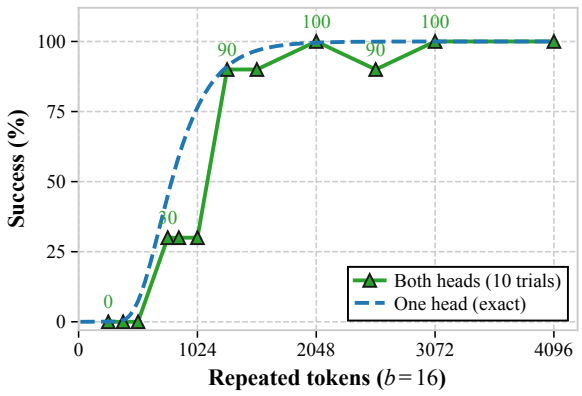


Figure 2. Calibration versus probe length for $b = 16$. The empirical series requires full recovery of both heads; the dashed series is the exact one-head occupancy probability. Ten trials are used at each empirical budget.

coupon-collector scale rather than estimating a deployment success curve.

5.3. RQ2: content recovery

Across the twelve configurations in Table II, the attack obtains 263,029 hits in 263,904 head–token opportunities, or 99.668%. Per-configuration recovery ranges from 99.318% on Phi-3-mini to 100% on six configurations. Fresh-secret control recovery ranges from 0 to 0.625%: ten configurations have zero hits, while the $b = 8$ and $b = 32$ configurations each have one hit. These controls support the role of the recovered secrets in the observed recovery.

We separately exercise two implementation variants whose records do not share the main table's denominator. In the fused path, an isolated attacker instance uses the original public model and recovered transform; it does not read the defender's fused weights. Recovery is 100% on each of the English, Chinese, and Python targets tested. In the optional `need_ratio` path (S and M ratio 1.25), the variant-specific recovery succeeds on all five target texts and both permutation seeds. These results establish reachability for the tested variants, not coverage of their full parameter spaces.

Where the misses occur. Phi-3-mini contributes 768 of the 875 misses and the largest denominator, 112,640, through its 32 KV heads; its recovery is 99.318%. Thus the aggregate weights both head counts and tokenized lengths, rather than models equally.

Table III regroups the same counts by fixed target text, adding no samples. Configurations, heads, and permutation seeds repeatedly evaluate each text. Different vocabularies, lengths, and model weights prevent interpreting these rows as language- or application-level recovery rates.

The counters cannot attribute misses to numerical error, marker selection, or dictionary ambiguity. High parameter cosine similarity alone does not guarantee exact recovery at small dictionary margins; Section 3 states the conditional dependency without assigning causes to unretained errors.

Language-prior ordering follow-up. The algebraic decoder returns block multisets. A separate Qwen2.5-0.5B follow-up orders these using public-model beam search (width 16), preserving token multiplicity and conditioning on at most 96 tokens of its own reconstructed context, starting from a model start token rather than a victim prefix. It contains 600 synthetic records in three categories and four nominal length groups, 50 per cell. The groups are generation minima: 109 records exceed their labels, and the 600 records contain 581 distinct evaluated token spans. Table IV therefore reports descriptive means by nominal minimum, not an exact-length benchmark.

Mean per-record position accuracy is 22.3%, versus 7.4% after randomizing each recovered block and 18.9% when every block restarts without preceding reconstructed context. The logs record 18/600 exact spans and 108/2,360 exact blocks. Replacing recovered bags with truth bags gives identical recorded metric vectors in all 600 records. Unit conservation in that oracle condition indirectly supports whole-span bag agreement under the inspected code; pooled conservation does not certify each block's bag. Raw predictions and recovered counters were not retained.

On 100 random-token records, ordering yields 5.5% versus 7.1% randomized ordering (Figure 6). This supports a role for language predictability, without proving that no ordering information exists. A further 100 direction-reversal pairs have 19.0%/18.5% position accuracy in their two groups and no exact spans, but the groups' scores are not direction-classification accuracy. Every pair has different blockwise bags despite equal global bags; these pairs cannot establish zero factual-direction leakage. Appendix E documents the retrospective protocol and retained-record limitations. This study supersedes the defective exploratory ordering estimates.

5.4. RQ3: entry point and secret lifetime

Official entry point. We construct the directory layout expected by the authors' command-line pipeline, invoke their configuration generator and obfuscator, and attack the protected files they produce. The targets are 16 English conversations selected from a public LMSYS-Chat-1M mirror [9]. The defender's magnitude thresholds are computed from the first eight target conversations; this setup is not an independent calibration/test split. Target plaintext is not supplied to the recovery routine. The saved per-file rates are 100% for all 16 files. This record supports a controlled file-format test, not a production-service or remote-access claim. The historical harness reused an output directory and did not enforce subprocess success before reading files, so the saved summary alone does not certify a fresh end-to-end execution.

Multiple secret epochs. Figure 3 summarizes three models and five independently generated configurations per model. All heads calibrate in all 15 model–epoch cells.

TABLE II. MAIN-SUITE RECOVERY FROM FIRST-LAYER VALUE CACHES. "PROBE" IS THE COMPLETE TOKENIZER OUTPUT LENGTH; C/N IS THE RAW MULTISET HIT COUNT. CELL COLORS ENCODE PERCENTAGES IN FOUR EQUAL BANDS FROM GREEN (LOW) TO RED (HIGH); COLOR DOES NOT INDICATE STATISTICAL SIGNIFICANCE.

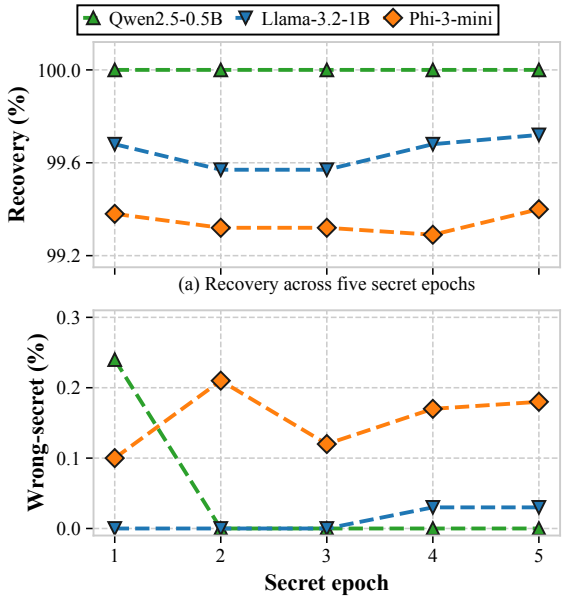
Model	Configuration				Token recovery		Fresh secret
	b	Heads	d	Probe	C/N	Rec. (%)	Control
Llama-3.2-1B	16	8	64	3,201	20,192/20,224	99.84	0/176
OLMo-2-1B	16	16	128	3,200	45,043/45,056	99.97	0/176
Phi-3-mini	16	32	96	4,080	111,872/112,640	99.32	0/208
Qwen2.5-0.5B	16	2	64	3,200	4,800/4,800	100.00	0/176
	32	2	64	20,250	4,608/4,608	100.00	1/160
	8	2	64	1,280	4,896/4,896	100.00	1/176
Qwen2.5-1.5B	16	2	128	3,200	4,768/4,800	99.33	0/176
Qwen2.5-3B	16	2	128	3,200	4,800/4,800	100.00	0/176
Qwen2.5-7B	16	4	128	3,200	9,578/9,600	99.77	0/176
Qwen3-0.6B	16	8	128	3,200	19,200/19,200	100.00	0/176
Qwen3-1.7B	16	8	128	3,200	19,200/19,200	100.00	0/176
TinyLlama-1.1B	16	4	64	2,033	14,072/14,080	99.94	0/208
				Total	**263,029/263,904**	**99.668**	–

TABLE III. THE MAIN-SUITE COUNTS REGROUPED BY FIXED TARGET TEXT. THESE ARE THE SAME CORRELATED EVALUATION OPPORTUNITIES AS TABLE II, NOT ADDITIONAL EXPERIMENTS.

Target text	Matches / opportunities	Misses
English prose	33,795 / 33,920	125
Python code	58,587 / 58,784	197
Chinese prose	80,517 / 80,768	251
Dialogue	37,928 / 38,048	120
Mathematical prose	52,202 / 52,384	182
Total	**263,029 / 263,904**	**875**

TABLE IV. ORDERING FOLLOW-UP: MEAN POSITION ACCURACY (%) BY NOMINAL GENERATION MINIMUM, WITH 50 RECORDS PER CELL. ACTUAL SPANS EXCEED THE MINIMUM IN 109/600 RECORDS.

Type	Nominal minimum blocks			
	1	2	4	8
Conversation	45.2	30.2	25.0	17.5
Narrative	37.6	24.5	23.6	17.6
Structured	8.8	15.0	11.6	11.0



(a) Recovery across five secret epochs

(b) Fresh-secret control

Figure 3. Secret-epoch stability on three models. Every epoch calibrates all heads. The y-axis in (a) is deliberately zoomed; (b) uses demixers from the wrong configuration. The conversations are not an exclusive held-out set.

Mean recovery across epoch means is 100% for Qwen2.5-0.5B, 99.64% for Llama-3.2-1B, and 99.34% for Phi-3-mini; the lowest model-level epoch mean is 99.29%. Wrong-secret controls average 0.05%, 0.01%, and 0.16%, respectively. The same ten public conversations recur across epochs and overlap earlier public-data experiments, so we treat this as secret-epoch stability, not held-out-session generalization. The result file retains rounded session rates rather than raw counts.

Lifecycle simulation. In a controlled Qwen2.5-0.5B service simulation, one calibration request is followed by ten synthetic requests from different application domains. Recovery is 100% on all ten. Replacing the complete secret configuration reduces a stale demixer to 0.78%; one new calibration probe restores 100% on the measured request. This result identifies the relevant unit of exposure: a shared secret epoch. It does not estimate tenant co-location prevalence in real deployments.

Recalibration after rotation uses a new shared epoch; it does not show that an old demixer survives rotation. The request-refresh control separately tests transfer without shared row secrets.

5.5. RQ4: bounded privacy consequence

We evaluate closed-set identification on 30 synthetic cases. For each of six value types—SSN-like strings, card numbers, passwords, medical record numbers, names, and addresses—we generate a fixed list of 100 candidates and use the first five as truths. A target value is embedded in neutral context. The attacker aggregates recovered tokens from all complete blocks and all heads; it is not told which block contains the value. A candidate is scored by the fraction of its unique tokens present in the recovered multiset.

Let g be the number of candidates with a strictly higher score than the truth and m the size of the truth's tie group. The truth attains the maximum in all 30 cases ($g = 0$); it is the unique maximum in 21/30. Under uniform random tie-breaking within the recorded group, expected top-1, top-5, and top-10 success are 77.36%, 90.95%, and 95.24%, respectively. Uniform guessing from 100 candidates gives 1%, 5%, and 10%. Figure 4 shows the top-1 result by value type.

The expected top-k contribution of a case is

$$p_k(g, m) = \min\{1, \max\{0, (k - g)/m\}\}. \qquad (14)$$

There are g candidates ahead of the tie group and m equally likely positions within it. If the remaining top-k slots cover the entire group, success is one; otherwise the fraction of covered positions determines the expectation. Thus $g = 0$ means that the truth is not outranked, whereas unique identification additionally requires $m = 1$. We average these per-case expectations over the 30 retained cases, rather than treating tied maxima as deterministic successes.

The candidate score tests token membership, not order or a verified field span. A candidate can share tokens with the surrounding context or with another candidate. Even perfect recovery of a target multiset can therefore leave several values tied. Conversely, identifying a candidate from a supplied list does not imply that the method can generate that value without candidate knowledge. These distinctions explain why the privacy outcome and the main multiset metric are complementary rather than interchangeable.

This experiment is evidence of consequence, not a population estimate. Truths are the first five generated candidates rather than uniform random indices; the displayed 1% line is therefore a uniform-guess reference, not a matched optimal prior. Only the truth score and (g, m) summaries were retained, not all candidate scores or recovered multisets. An "entity complete" flag used a token-set approximation rather than a verified character span. We consequently do not claim unconstrained credential recovery, stable per-type rates, or complete-field reconstruction.

5.6. Shortening the secret lifetime

The mechanism suggests refreshing the reused row transform S and marker a while leaving the fused column transform M static. In one Qwen2.5-0.5B experiment, the saved same-epoch recovery rate is 97.44%, compared with

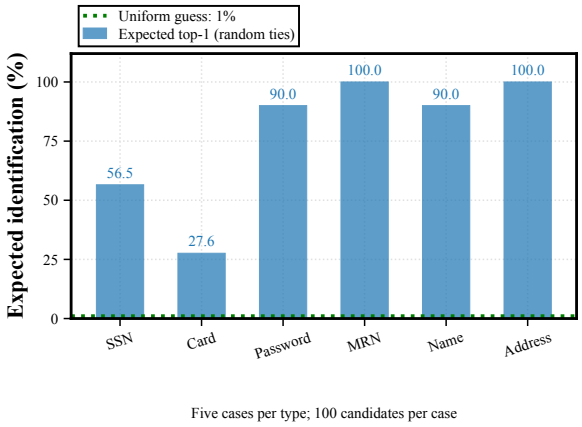


Figure 4. Closed-set identification in the 30-case synthetic study. Each bar contains five cases. Ties are broken uniformly in expectation.

0% when a demixer calibrated for request A is applied after S, a are regenerated for request B. The result JSON retains rounded rates rather than raw counts for this comparison. Generating the corresponding matrices for the full model takes 2.9 ms in a CPU microbenchmark.

This is a mitigation candidate, not a validated repair. The saved round-trip check does not establish full K/V functional equivalence, and the test does not cover adaptive within-request calibration, shared prefixes, multi-turn identity, device-transfer cost, or end-to-end serving latency. The defensible conclusion is narrower: request-specific S, a prevents the *existing* cross-request demixer from transferring in the measured setting.

5.7. Answers to the research questions

The observed state counts and parameter errors support the predicted marker orbit (RQ1). Recovery is high across the tested models and configurations, while fresh-secret control recovery remains between 0 and 0.625% (RQ2). The attack consumes the official protected-cache output and remains effective across generated secret epochs, provided calibration and target caches share the relevant secrets (RQ3). The synthetic closed-set study exceeds the uniform-guess reference, with candidate selection and incomplete traces limiting its interpretation. Shortening the row-secret lifetime interrupts the measured transfer, but functional correctness and general security remain unverified (RQ4).

6. Discussion

6.1. The security lesson: observable entropy

The failure is not that the implementation forgets to sample a permutation. It samples one for each block. The failure is that, for a reachable input symmetry, many sampled permutations become observationally equivalent. The security-relevant quantity is the orbit of the transformed

object, not the cardinality of the randomizer's support. A $b!$ implementation choice can therefore contribute only b distinguishable calibration states.

This distinction is broader than KV-Cloak, but the attack is not a theorem about all linear obfuscation. A designer must ask which transformations survive after quotienting by input stabilizers, which fixed structure remains across observations, and whether recovery metadata becomes an oracle. In the tested construction, the single-row marker supports the implementation's decoding procedure and also labels states that expose the columns of S. Different marker or mixing structures can change the orbit and the identification problem.

6.2. Mitigation implications

Three principles follow. First, do not reuse a transform merely because another component is refreshed: the refreshed component may vanish under a chosen input. Second, bind secrets to the smallest practical identity—request, tenant, or cache object—and specify how shared prefixes and multi-turn sessions inherit that identity. Third, evaluate functional recovery and security together. Replacing a sparse marker with a denser object can defeat this decoder while also breaking the intended inverse; failure of the current attack is not evidence of a correct defense.

Request-specific S, a is the simplest intervention suggested by our measurements. It changes the attacker's problem from amortizing one calibration over other requests to calibrating within the victim's secret scope. Yet the correct engineering choice may instead be authenticated encryption when caches are only stored or transported and need not be manipulated while protected. Obfuscation should be justified by an operation that conventional cryptography cannot efficiently support, not by a general claim that transformed vectors are unintelligible.

Secret lifetime must also be considered together with cache lifetime. A request-scoped transform requires the legitimate decoder to retain the association between a cache object and the parameters that protect it. Shared prefixes or a multi-turn session may span several requests, so the term *per request* does not by itself specify a complete integration policy. Our parameter-generation microbenchmark measures neither that state management nor re-encoding of existing cache objects. The observed transfer failure is consequently a mechanism-level intervention whose integration remains to be validated.

The distinction between functional and security validation is particularly important here. A cache transform can interrupt our numerical recovery while also changing legitimate attention outputs. Establishing a usable repair would require checking corresponding K/V rows, masks, and continued inference under the proposed secret-scope policy, followed by attacks matched to that policy. The present work does not claim those checks have been completed. Its contribution is the demonstrated failure of the reused-secret configuration and the bounded evidence about which component refresh interrupts the measured transfer.

6.3. A measured defense prototype

As a follow-up design exercise, we implemented VeilKV-Plus, a two-profile prototype that removes the beacon, derives block transforms from HKDF-domain-separated material, authenticates cache blocks, and separates public-shareable prefixes from private tails. The TEE profile protects serialized K/V bytes with standard encrypt-then-MAC processing and performs key release through an attestation stub; the arithmetic profile retains a reversible floating-point mixing layer for settings without byte-level decryption. We report this as a defense candidate under the simulated TEE boundary, not as a proof of a deployed TEE system. The v4 run covers Qwen2.5-0.5B and Llama-3.2-1B, retains run-1 attack and performance evidence, and records all stage outputs in the artifact.

The prefix result is now defined at the artifact-manager boundary. A public prefix artifact is authenticated with block-scoped metadata and can be reused across requests; the private tail starts from the terminal tag of the reused prefix chain. In the v4 paged harness, cross-request prefix byte identity was $1/1$, chain-aware prefix-to-tail decoding matched the unprotected BF16 baseline (K error 0.49898, V error 0.0312), and greedy decoding matched for 20 steps. Wrong terminal tags, wrong principals, block swaps, truncation, eviction/reload, relocation, and page substitution were detected. The guarantee concerns reuse of one stored artifact; independently recomputed forward caches can still differ because deeper-layer floating-point reductions depend on the request shape.

The v4 integrity battery detected all 12 tested manipulations, including replay after epoch changes, block substitution, block swapping, truncation, AAD changes, and cross-tenant replacement. Private closed-set dictionary attempts produced $0/100$ hits for both evaluated models, while the public-prefix equality oracle remains an intentional policy-bound leakage. The persistent nonce store rejected reuse after a simulated service restart. Attestation remains a stub, so the result establishes the cache protocol and its failure handling rather than the security of a particular hardware TEE.

The TEE profile had protect/recover medians of 0.1159/0.0987 seconds per 1,000 tokens in the CPU benchmark inherited by v4, compared with 0.2238/0.4142 seconds for the KV-Cloak comparison. These are approximately $1.93\times$ and $4.20\times$ reductions in the measured operation time against the benchmark's `fused=False` baseline, not the published weight-fused serving path. The benchmark measures cache protection and recovery; it excludes hardware attestation and does not establish an end-to-end serving speedup.

6.4. Baseline comparison and supported advantages

Table V separates the intervention suggested by the attack from the engineering properties tested by the defense prototype. Request-specific S, a refresh interrupts the

measured cross-request transfer, but does not supply the integrity, replay handling, or prefix-to-tail integration evaluated in VeilKV-Plus. The TEE profile addresses these additional gaps by combining standard byte protection with authenticated cache-object metadata, persistent nonce freshness, and an explicit distinction between public prefix artifacts and private tails. The cryptographic primitives are established tools; the prototype's evidence concerns their cache-protocol integration under the simulated trust boundary.

A no-sharing authenticated-encryption baseline provides a closer comparison for the cost of this integration. Its median protection time is 0.1091 seconds per 1,000 tokens, versus 0.1159 for the TEE profile: the latter pays approximately 6.2% more for protection in this benchmark. The saved record does not provide a corresponding no-sharing recovery measurement. Thus the supported advantage over that baseline is the demonstrated reuse of one policy-public prefix artifact with an authenticated private continuation, rather than faster encryption or stronger cryptographic primitives. This does not establish that other encryption-based systems cannot support sharing. Our reuse test comprises one cross-request artifact pair and a paged harness, not a production hit-rate or throughput study.

The public/private distinction is essential to this advantage. A public prefix's equality remains observable by design, and its reuse cannot be advertised as confidentiality for arbitrary low-entropy private prefixes. Private-tail dictionary attempts instead exercise principal- and nonce-scoped protection under the simulated TEE assumption. The stored-artifact test also differs from independently recomputing a prefix: deeper-layer floating-point reductions can change with request shape. Those differences are a reproducibility constraint, not a security guarantee.

6.5. Negative result for the arithmetic profile

The subsequent optimization study resolves the earlier low-power spectral-leakage concern. Its retained per-block records contain 100 blocks per model and 100 candidates per identification task. Gram-based top-1 identification succeeds on 23/100 Qwen2.5-0.5B blocks and 38/100 Llama-3.2-1B blocks; the recorded 95% Wilson intervals are [15.84%,32.15%] and [29.10%,47.79%]. Whitening succeeds on 18/100 and 6/100, respectively. The uniform candidate baseline is 1%, while the sampled random controls obtain 0/100 and 2/100. These observations support a closed-set identification leak in the arithmetic profile; they do not measure exact prompt reconstruction or deployment-wide leakage rates.

Optimization improves arithmetic protection/recovery means to 0.7325/0.5789 seconds per 1,000 tokens in the final longer-input benchmark (464 actual tokens). The earlier approximately 11.4-second figure came from a short-input benchmark and cannot serve as the same-input baseline for a speedup claim. In the final benchmark, the recorded KV-Cloak means are 0.0596/0.0969 seconds per 1,000 tokens. Performance therefore remains unfavorable, and the identification result independently excludes the arithmetic profile from confidentiality claims. Improving numerical reversibility or runtime does not remove its observed leakage. The TEE-profile evidence remains scoped to byte protection under a simulated, static master key and an unimplemented hardware attestation protocol.

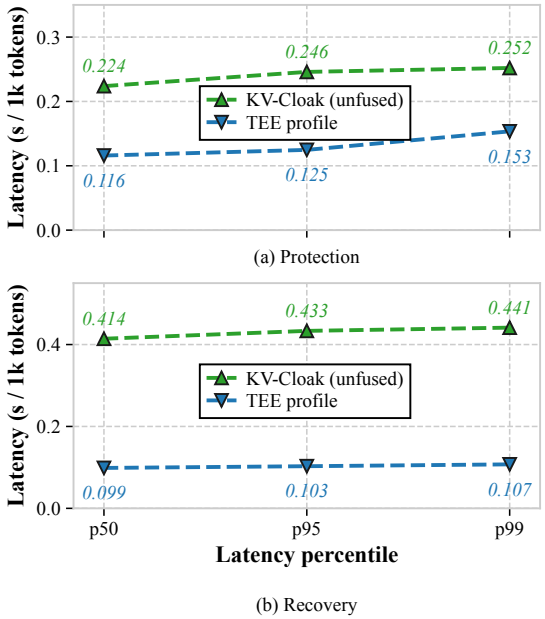


Figure 5. Latency percentiles from the frozen VeilKV-Plus v4 CPU benchmark. The TEE profile remains below the unfused KV-Cloak comparison for both protection and recovery; the plot reports the measured profile only and does not encode a security guarantee.

Figure 5 reports the full latency percentiles for the two byte-oriented profiles. It makes the operational tradeoff visible without combining latency with the separate integrity and confidentiality tests.

6.6. Limitations and external validity

Our attacker begins after protected-cache observation is available. We do not show a remote API path to another tenant's cache, measure the prevalence of shared secret epochs, or bypass access control. The study concerns the first Value layer and the tested public implementation. Deeper states, Key recovery, architectures with position-dependent Value inputs, arbitrary orthogonal M, and marker patterns outside the analyzed orbit require separate arguments.

The direct output is unordered within a block. Language-prior sorting gives partial sequence estimates, while the direction-pair diagnostic does not control blockwise bags. Neither observation establishes zero factual leakage or general verbatim reconstruction. The synthetic value study uses one model, one secret configuration, 30 cases, and an artificial candidate set. Its incomplete retained traces rule out post-hoc analyses that would require all scores or recovered

TABLE V. SCOPE OF THE MEASURED INTERVENTIONS. UNMEASURED PROPERTIES ARE LEFT EXPLICIT. PUBLIC PREFIX REUSE PERMITS EQUALITY LEAKAGE; THE TEE BOUNDARY IS SIMULATED.

Property	**KV-Cloak comparison**	**Request-specific S, a**	**VeilKV-Plus TEE profile**
Secret scope	Reused structural secrets in the tested configuration	Refresh S, a; retain M	Derived material scoped to cache metadata and private principal/nonce
Confidentiality evidence	First-layer Value multiset recovery in the tested implementation	Measured transfer falls to 0%; general security unestablished	Private dictionary: 0/100 per model; hardware key release unimplemented
Integrity and replay	Not evaluated as a guarantee in this comparison	Not evaluated	12/12 tested manipulations detected; nonce replay rejected after restart
Prefix and tail integration	No corresponding chain-aware repair test	Not evaluated	Stored public prefix reused; tail linked to its terminal tag; wrong joins rejected
Functional boundary	Official cache-output recovery evaluated	Full K/V correctness unvalidated	FP32 checks and no error beyond the unprotected BF16 storage baseline

blocks. The official-entry and multi-epoch studies reuse public conversations; neither supports a claim about an unseen deployment distribution.

The main 99.668% total weights models with more KV heads and texts with more complete blocks more heavily. It is an audit total, not an estimate over a defined population of users. The model and checkpoint breadth nevertheless tests whether the mechanism is tied to one tokenizer or hidden dimension. Claims about prevalence, per-user harm probability, and production cost remain outside the evidence.

6.7. Ethics, disclosure, and release

All experiments ran locally on public model checkpoints, synthetic text, or a publicly released conversation dataset [9]. We did not recruit or interact with human participants, query a live third-party service, access private caches, or recover non-public user data. No separate human-subjects protocol was submitted because the study involved no participant interaction or new private-data collection. The public conversation subset is used only to exercise file-format and recovery behavior; the manuscript does not reproduce conversation text or identifiers.

The attack is dual use because cache-read access could be converted into content leakage. The authors of KV-Cloak were notified of this vulnerability via a GitHub issue on September 28, 2026. That initial notice requested a private reporting channel and did not include technical details. The accompanying research artifact contains local experiment scripts and frozen results; its publication makes technical details accessible. The initial notification does not establish maintainer acknowledgment, delivery of a detailed private report, or an agreed release timeline.

The source snapshot, frozen result records and their hashes, and figure/table generation scripts are provided through the anonymous artifact at https://anonymous.4open.science/r/fresh-shuffles-artifact-ED8D/. The artifact distinguishes exploratory results from records supporting the paper's claims. Any change to a frozen result or denominator requires regenerating the evidence manifest and manuscript assets.

7. Related Work

7.1. KV-cache and internal-state privacy

KV caches are central to efficient autoregressive serving; PagedAttention makes fixed-size cache blocks a first-class memory-management unit [1]. The same states encode prompt information. Prior work reconstructs text from embeddings and internal activations through optimization, generation, or dictionary search [13], [14], [15]. KV-Cloak demonstrates direct, collision, and injection attacks against unprotected caches, then proposes reversible cache obfuscation [2]. We analyze that proposed protection rather than repeating its attacks on plaintext caches.

7.2. Attacks on permutation and obfuscation defenses

Hidden No More recovers prompts from permuted hidden states and challenges proofs based on permutation hardness [10]. Saini et al. exploit reuse of precomputed noise bases in partial-TEE systems using chosen internal vectors, recovering secret permutations and weights [11]. Collapse gives a primitive-level analysis of weight-obfuscation schemes and attacks compositions that retain algebraic structure across refreshed views [12]. These works establish that permutation, linear mixing, and refresh are not independently persuasive security arguments.

Our observation and target differ. The attacker supplies legal text rather than arbitrary hidden vectors; the protected object is a runtime Value cache; an unknown dense row transform remains; and a new row permutation is drawn per block. Repeated text makes the data term invariant, while

TABLE VI. CLOSEST ATTACK MECHANISMS. "REFRESH" DESCRIBES THE COMPONENT RELEVANT TO THE CITED ATTACK, NOT EVERY RANDOMNESS SOURCE IN THE SYSTEM.

Work	Protected asset	Attacker input/view	Refreshed structure	Distinguishing step
Hidden No More [10]	hidden states	public model; permuted states	per-scheme permutation	candidate distances reverse the hidden permutation
Partial TEE [11]	weights and activations	ENCRYPT queries incl. zero and basis vectors	fixed precomputed noise basis	cancel noise subspace; recover permutation and weights
Collapse [12]	obfuscated weights	multiple exposed weight views	low-rank mask, scaling, permutation per epoch	primitive decomposition, rank/overlap tests, alignment
This work	runtime Value cache	normal text; protected cache	row permutation per block	input symmetry collapses orbit; marker differences recover equivalent demixer

a sparse recovery marker turns the fresh permutations into a finite-state signal. Table VI makes this mechanism-level boundary explicit.

7.3. Confidential inference and refreshed mixing

Confidential inference spans secure computation, trusted execution, differential privacy, obfuscation, and hybrids; the recent SoK by Yaluhin compares these families and emphasizes their distinct trust and deployment costs [16]. GELO refreshes invertible mixing per batch and explicitly studies Gram leakage, blind-source separation, and shield vectors [17]. ConjFormer instead changes the architecture to be orthogonally equivariant, permitting inference in a rotated basis [18]. These designs reinforce the importance of refresh scope and of testing invariants exposed by structured transforms.

We do not propose a new general obfuscation system. Our mitigation result is intentionally modest: shortening the lifetime of two reused components prevents the current demixer from transferring in one setting. This preserves the paper's contribution as an attack and analysis, and avoids treating failure of a finite attack battery as a security proof.

7.4. Positioning relative to the original CPA

The KV-Cloak paper already considers chosen-plaintext attacks on simpler linear constructions and introduces the marker and per-block permutation as part of the hardened design [2]. Repeated or low-rank inputs and public-model dictionaries are therefore not, by themselves, new contributions. The difference is retaining the complete published transform and using the marker's *post-permutation orbit* to recover the reusable structure despite a fresh P for every block. Our evaluation accordingly includes the official entry point, fused mode with an isolated public dictionary, fresh-secret controls, and secret-lifetime tests.

This distinction can be stated at the level of the observed matrices. The original paper's naive construction has a fixed linear relation between plaintext differences and ciphertext differences. Its hardened construction inserts a new block permutation, so the original relation cannot simply be assumed to align two arbitrary observations. Our calibration input instead fixes the data term under all row permutations. The variation that remains comes from the marker. The analysis studies this remaining variation directly, without assuming that the sampled permutations are equal or that their labels are known.

Collapse studies exposed weight views with additive masks, scaling, and refreshed permutations [12]. Our protected caches additionally contain dense row mixing that must be identified before dictionary decoding. Repeated-input marker states supply that signal. The distinction concerns the observed matrix structure and input capability; refreshing randomness and linear-algebraic recovery are themselves established topics.

8. Conclusion

Fresh randomness is useful only to the extent that it creates fresh observable uncertainty. In KV-Cloak, a legal repeated-token request makes the first-layer Value term invariant to row permutations. The remaining marker visits a small orbit whose rank-one differences identify an equivalent demixer. Across ten models and twelve configurations, the attack recovers 263,029 of 263,904 head–token opportunities, while wrong-secret controls remain between 0 and 0.625%. The attack also consumes the official protected-cache output and transfers across requests within a secret epoch.

The attack establishes blockwise token-multiset recovery under cache observation. A 600-record follow-up logs 22.3% mean position accuracy using a public language prior; this does not establish general ordered or factual reconstruction. Deployment prevalence and a complete repair remain unestablished. A limited closed-set study illustrates the privacy implications, while request-specific refresh blocks the tested cross-request transfer without proving general security. The broader design lesson is to analyze the observable orbit induced jointly by randomness and reachable inputs. Counting random choices without accounting for this orbit can overstate protection.

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Appendix A. Reproducibility Details

A.1. Model revision inventory

Table VII records local Hugging Face cache revisions observed during artifact preparation for the checkpoint identifiers in the main suite. These are retrospective provenance records: the experiment outputs did not bind each run to a revision hash. Qwen2.5-0.5B is reused for the block-size variants and the focused studies.

The official-entry script used the public mirror `AarushSah/lmsys-chat-1m`. Its local cache revision observed during artifact preparation is `c9fce24f9623e40750a061581bdd3a2d1243ff79`; the saved result contains truncated hashes and accuracies, not conversation text or a run-bound dataset revision.

A.2. Evidence and generated assets

The artifact separates model-running code from manuscript generation. JSON records under `paper/evidence/` are copied from the

TABLE VII. CHECKPOINT IDENTIFIERS AND CACHE REVISIONS OBSERVED DURING ARTIFACT PREPARATION. THESE REVISIONS WERE NOT RECORDED IN PER-RUN MANIFESTS.

Checkpoint	Revision
Qwen/Qwen2.5-0.5B	060db6499f32faf8b98477b0a26969ef7d8b9987
Qwen/Qwen2.5-1.5B	8faed761d45a263340a0528343f099c05c9a4323
Qwen/Qwen2.5-3B	3aab1f1954e9cc14eb9509a215f9e5ca08227a9b
Qwen/Qwen2.5-7B	d149729398750b98c0af14eb82c78cfe92750796
Qwen/Qwen3-0.6B	c1899de289a04d12100db370d81485cdf75e47ca
Qwen/Qwen3-1.7B	70d244cc86ccca08cf5af4e1e306ecf908b1ad5e
unsloth/Llama-3.2-1B	9535bd9b1d1dea6acafbdc4813b728796aeb28da
allenai/OLMo-2-0425-1B	a1847dff35000b4271fa70afc5db10fd29fedbdf
microsoft/Phi-3-mini-4k-instruct	f39ac1d28e925b323eae81227eaba4464caced4e
TinyLlama/TinyLlama-1.1B-Chat-v1.0	fe8a4ea1ffedaf415f4da2f062534de366a451e6

frozen experiment results and listed with SHA-256 hashes in `manifest.json`. The script `paper/scripts/generate_assets.py` reads only those records, regenerates all manuscript figures and an audited copy of the main-table rows, recomputes the reported privacy tie statistics, and asserts the principal denominator 263,029/263,904. The submission checker verifies that every generated row is present verbatim in the manuscript source. Neither script makes a model or network call.

The original runs did not persist a complete contemporaneous Python dependency lock. The artifact preserves source snapshots, a retrospective model-revision inventory, result records, and manuscript-generation requirements. It does not establish bit-identical model reruns. This limitation does not affect hash verification or regeneration of paper assets from the frozen records.

A.3. Architecture and software provenance

The Qwen3 and OLMo 2 reports provide the generation-specific background for those checkpoints [19], [20]. The scripts load checkpoints and tokenizers through Transformers [21] and perform tensor operations in PyTorch [22]. These references identify the software and model families; they do not substitute for a historical dependency lock or prove which revision produced an earlier result.

The distinction between query heads and KV heads follows the multi-head and grouped-query attention setting [23], [24]. Our denominators use the recorded KV-head count, since those are the Value representations being scored. The normalization operator in Equation (1) is taken from the loaded model, including RMSNorm where applicable [25]. Applying the raw embedding table without the relevant normalization and projection would evaluate a different dictionary.

A.4. Regeneration contract

From the repository root, install `requirements-paper.txt` in a Python 3.12 environment. Run `paper/scripts/generate_assets.py` to regenerate the principal figures and audited main-table rows; `paper/scripts/generate\_order\_assets.py` generates the ordering diagnostics. Run `paper/scripts/generate_supplement.py` to regenerate the appendix tables; its `-check` mode compares the checked-in tables with freshly derived text. These commands consume saved JSON/JSONL only.

Compile `paper/main.tex` with Tectonic, placing output in `paper/build/`. The checks in `paper/scripts/check_submission.py` verify the result-copy hashes, page dimensions, appendix marker, and main-table rows. The separate `paper/scripts/audit_final.py` recomputes main counts and privacy expectations, checks the text-level breakdown, and inspects the PDF's actual fonts and artifact hyperlink. All table generators use deterministic ordering. PDF binary hashes may nevertheless differ after rebuilding because compilation metadata can change.

This contract distinguishes three operations. Hash verification detects whether a supplied result differs from its recorded copy. Re-analysis checks arithmetic and presentation against that copy. Re-running a model would regenerate the cache observations and scores themselves. Only the first two operations are covered by the documented manuscript-generation commands. They cannot reconstruct candidate scores, input traces, or runtime provenance that were not saved.

Appendix B. Supplemental Measurement Details

B.1. Reconstructing the main denominator

Let $L_{c,t}$ be the tokenized length of target text t in configuration c, with block size b_c and H_c calibrated heads. The two recorded permutation seeds contribute

$$N_c = 2H_c \sum_t b_c \lfloor L_{c,t}/b_c \rfloor . \tag{15}$$

This identity applies to the present records because all configured heads calibrated. If a head failed, a sum over

TABLE VIII. CLOSED-SET CASES BY TYPE. TOP-k PERCENTAGES USE UNIFORM TIE-BREAKING FROM RETAINED g, m INTEGERS. FOUR EQUAL 25-PERCENTAGE-POINT COLOR BANDS RUN FROM GREEN (LOW) TO RED (HIGH). EACH ROW CONTAINS FIVE CASES UNDER ONE KEY.

Type	Unique	Max. m	Top-1	Top-5	Top-10
SSN-like	2/5	13	56.54	87.69	95.38
Card-like	1/5	20	27.61	58.03	76.05
Password	4/5	2	90.00	100.00	100.00
MRN	5/5	1	100.00	100.00	100.00
Name	4/5	2	90.00	100.00	100.00
Address	5/5	1	100.00	100.00	100.00

successful heads would instead be a conditional metric and would require an explicit coverage report.

For Qwen2.5-0.5B with $b = 16$, the saved complete-text lengths are 176, 288, 272, 192, and 272 tokens. Their sum is 1,200, and two heads under two seeds produce 4,800 opportunities. The fresh-secret control uses only the 176-token English text and head 0, producing 176 opportunities. Thus its denominator is deliberately smaller and is not interchangeable with the primary result.

The summed metric is a frequency over recorded opportunities. The same text appears across seeds and models, and its tokens appear in multiple heads. Neither the 263,904 denominator nor the five text categories supplies an independent user sample. The per-model and per-text tables are two views of the same records, not independent replications.

B.2. Per-type candidate statistics

Table VIII expands the closed-set study's tie summaries. Every type has five cases with $g = 0$. The largest tie group contains 20 candidates, illustrating why attaining the highest score is weaker than unique identification.

For example, a recorded SSN-like case has $g = 0, m = 13$. Equation (14) assigns it expected success $1/13$, $5/13$, and $10/13$ at top-1, top-5, and top-10. A case with $m = 1$ contributes one to all three metrics. Applying the upper cap of one before averaging is essential: without it, a unique maximum would incorrectly contribute five to top-5 success.

The retained completeness flag marks 15 cases as complete and 15 otherwise. On the marked-complete subset, unique maxima occur in 8/15 cases and expected top-1 is 61.38%. This subset is reported solely to expose the flag's behavior. Because it compares isolated entity-token membership rather than verified contextual spans, it cannot establish which fields were fully present or truncated. The lower subset rate is not evidence that truncation improves identification.

Truth selection is also fixed: the first five candidates generated for each type become targets. Uniform guessing over all 100 candidates is a reference calculation, not the strongest prior under every disclosure of the generation protocol. These restrictions, and the absence of the complete candidate-score vectors, preclude a population-level estimate of identification advantage.

Appendix C. Configuration and Epoch Diagnostics

Table IX reports the weakest recorded parameter diagnostics in each main configuration. Separation is the minimum across its heads; the S column is the minimum matched-column cosine; the w column is the largest relative error. These extrema summarize the saved diagnostics rather than new measurements. The ground-truth transform is used only to compute them.

The prefix lengths explain an otherwise misleading field in several legacy JSON files. Their `premise_rows_identical` check compares the entire input, including the prefix, whereas the calibrator slices to complete blocks inside the repeat run. A false whole-input flag therefore does not by itself refute the repeated-core premise. The recorded prefix and core-block counts must be interpreted together.

Table X gives every model–epoch cell underlying Figure 3. Epoch indices are displayed as 1–5, corresponding to stored indices 0–4. The mean is an arithmetic average of the ten saved session rates; the minimum is the weakest of those sessions. Control rates come from the fresh-secret comparison. They are rounded summaries, not reconstructed raw counts or confidence intervals.

The separation between calibration coverage and recovery remains necessary even though all heads succeed here. A complete state collection can still produce numerical parameter error, and a small error can alter a nearest-neighbor decision. Conversely, a high per-session average does not reveal whether its few mistakes share a marker hypothesis, a token identity, or a numerical cause. The retained summaries support the reported extrema but do not support a retrospective per-error attribution.

Appendix D. Details of the Algebraic Bounds

D.1. Marker multiplicities and the orbit

If the rows of a fixed marker matrix fall into equality classes of sizes $n_1, \dots, n_r$, permutations within each class leave the marker unchanged. Its stabilizer therefore has size $\prod_{\ell=1}^{r} n_\ell!$, yielding an orbit of $b!/\prod_\ell n_\ell!$. A single nonzero row and $b-1$ zero rows give b states. This count is algebraic; it does not by itself certify that states are distinguishable at the working precision or that a recovery algorithm identifies the mixing.

D.2. The sign test

The incorrect sign candidate in the identification proof gives $w' = -w - 2a/b$. Expanding the squared norm gives

$$\|w'\|^2 - \|w\|^2 = \frac{4}{b}\langle w, a\rangle + \frac{4}{b^2}\|a\|^2. \tag{16}$$

TABLE IX. MAIN-SUITE CALIBRATION DIAGNOSTICS DERIVED FROM SAVED HEAD RECORDS. PREFIX IS THE NUMBER OF LEADING TOKENS BEFORE THE REPEAT RUN; CORE BLOCKS EXCLUDE PREFIX-CONTAINING AND INCOMPLETE BLOCKS. ALL HEADS RECORD b CLUSTERS.

Model	b	Prefix	Core blocks	Min. separation	Min. S cosine	Max. w rel. err.
Llama-3.2-1B	16	1	199	910.5	0.999997	0.010575
OLMo-2-1B	16	0	200	518.7	0.999997	0.009411
Phi-3-mini	16	1	254	826.8	0.999996	0.035399
Qwen2.5-0.5B	16	0	200	1031.7	0.999997	0.008775
Qwen2.5-0.5B	32	0	632	1045.4	0.999997	0.005972
Qwen2.5-0.5B	8	0	160	1416.6	0.999997	0.011496
Qwen2.5-1.5B	16	0	200	689.8	0.999997	0.008220
Qwen2.5-3B	16	0	200	692.3	0.999997	0.006331
Qwen2.5-7B	16	0	200	524.2	0.999997	0.009426
Qwen3-0.6B	16	0	200	542.2	0.999997	0.007513
Qwen3-1.7B	16	0	200	500.2	0.999997	0.010269
TinyLlama-1.1B	16	2	126	1048.0	0.999997	0.010634

TABLE X. ALL 15 RECORDED MODEL–EPOCH CELLS. EVERY HEAD CALIBRATES. RECOVERY COLUMNS SUMMARIZE THE SAME TEN REUSED SESSIONS PER MODEL; CONTROL RATES ARE REPORTED SEPARATELY. PERCENTAGES INHERIT THE PRECISION OF THE SAVED RECORDS.

Model	Epoch	Heads	Probe	Mean (%)	Min. session (%)	Control (%)
Qwen2.5-0.5B	1	2/2	3200	100.00	100.00	0.24
	2	2/2	3200	100.00	100.00	0.00
	3	2/2	3200	100.00	100.00	0.00
	4	2/2	3200	100.00	100.00	0.00
	5	2/2	3200	100.00	100.00	0.00
Llama-3.2-1B	1	8/8	3201	99.68	99.41	0.00
	2	8/8	3201	99.57	99.12	0.00
	3	8/8	3201	99.57	99.22	0.00
	4	8/8	3201	99.68	99.61	0.03
	5	8/8	3201	99.72	99.51	0.03
Phi-3-mini	1	32/32	4080	99.38	99.14	0.10
	2	32/32	4080	99.32	99.17	0.21
	3	32/32	4080	99.32	99.15	0.12
	4	32/32	4080	99.29	99.18	0.17
	5	32/32	4080	99.40	99.26	0.18

The norm test therefore separates the two candidates except at the degeneracy stated in Theorem 2. This exact-arithmetic statement does not assign a tolerance to the finite-precision implementation. The measured diagnostics in Table IX supply that implementation's empirical context without converting small errors into a uniform guarantee.

D.3. Exact occupancy curve

For one head with independent uniform marker positions, let $q_n(k)$ be the probability of observing exactly k states after n blocks. Starting from $q_0(0) = 1$, the recurrence is

$$q_{n+1}(k) = \frac{k}{b} q_n(k) + \frac{b-k+1}{b} q_n(k-1), \qquad (17)$$

with out-of-range probabilities set to zero. The first term repeats an observed state; the second introduces a previously unseen one. The dashed curve in Figure 2 uses $q_n(b)$, calculated deterministically by the figure generator. It is not the legacy JSON field `coupon_pred_all`, and it is not the probability that every numerical recovery step succeeds for both heads.

D.4. What a functional permutation check must preserve

A complete-row permutation preserves each row as a vector. Sorting every feature column independently does not test that property. For instance,

$$X = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}, \qquad Y = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

have identical independently sorted columns and identical row-norm multisets, but different row-vector multisets. Neither diagnostic establishes a row permutation. For attention, corresponding K and V rows must additionally share the same permutation, with mask and position semantics handled consistently. This is why the retained S/a check does not establish full functional equivalence.

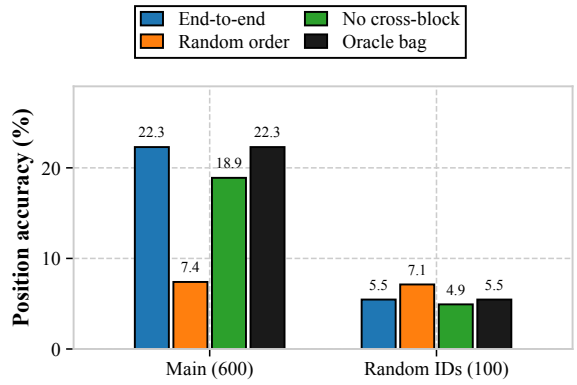


Figure 6. Recorded mean position accuracy in the ordering follow-up. Main contains 600 synthetic records (581 distinct evaluated spans); random IDs contains 100 records. These are descriptive per-record means, not independent token trials or direction-identification rates.

Appendix E. Ordering Follow-up and Scope

The delivered metrics contain 600 main and 300 boundary IDs, each with four conditions. Main prediction/cost streams contain 612 rows because of resumed duplicates; boundary prediction summaries retain 200 earlier short-input skips before 300 completed records. We analyze the unique metric IDs and preserve all physical rows. The files named predictions contain lengths and IDs, not output tokens; recovered counters were also not saved. Hashes freeze the delivered files and current source retrospectively. The separately supplied protocol is explicitly retrospective and is not a pre-run registration.

In `evaluate`, conservation compares the predicted sequence with recovered bags pooled over all blocks. The oracle condition orders truth bags but is scored against those recovered bags. Its unit conservation and equal lengths indirectly imply whole-span bag agreement under the inspected implementation; they cannot establish which block each token came from. Identical end-to-end/oracle metric vectors (600/600 main and 300/300 boundary) likewise do not certify identical sequences. Random-order means use the saved shuffled outputs; resumed RNG streams and the source's whole-span analytic reference prevent presenting that field as an exact blockwise null. No confidence interval treats repeated templates as independent observations.

A retrospective offline-tokenizer audit agrees with all 900 recorded lengths. It finds 109 nominal-length overshoots and 581 distinct main spans. The 100 boundary pairs comprise 68 distinct unordered token-sequence pairs. All 100 share a global bag, but zero share every block bag: tokens move between blocks, leaving the blockwise-content control required for an indistinguishability test absent. The reported 19.0%/18.5% values group position scores by input direction; there is no retained factual-direction classifier. Across both directions, randomized ordering gives 7.12%; the delivered derived file's 6.69% selects direction-a only. No exact boundary span is recorded (0/200), which does not establish absence of factual leakage.

The follow-up replaces the earlier multiplicity-losing exploratory estimates. Theorem 2 still identifies unordered columns of S; sequence estimates from language priors do not recover those original column labels or validate joint K/V attention semantics.